PORTFOLIO PROJECT · GENERATIVE AI

Using Generative AI to Improve **Knowledge Retrieval** for Customer Service Representatives

A RAG-powered assistant that helps CSRs find answers instantly during high-volume Annual Enrollment periods

Generative AI | RAG Architecture | Knowledge Management | LLM Integration | Vector Search

Tanima Roy
Knowledge Management & Documentation Professional

Why This Project Exists

During Annual Enrollment periods, Customer Service Representatives face a surge of benefit inquiries. The existing knowledge base creates critical bottlenecks:

Fragmented Knowledge Base

Articles are long, split across multiple pages, and hard to navigate quickly.

High Search Time

New CSRs spend 90–120 seconds per query searching — time better spent helping customers.

Increased Handle Time

Average Handle Time (AHT) rises during peak periods, slowing overall service delivery.

Inconsistent Answers

Different agents find different articles — leading to variable customer experiences.

What We Set Out to Achieve

Design a Generative AI–powered knowledge assistant that retrieves relevant articles instantly, generates concise answers, and provides source references from the knowledge base.

Search Time	Handle Time	CSR Confidence	Answer Quality
BEFORE	BEFORE	BEFORE	BEFORE
90–120 sec	Baseline	Low (new agents)	Variable
WITH GENAI	WITH GENAI	WITH GENAI	WITH GENAI
30–40 sec	**↓ 40%**	**High**	**Standardized**
Per query	Projected reduction	Guided responses	Consistent CX

RAG-Based GenAI Assistant

RETRIEVAL AUGMENTED GENERATION

A RAG pipeline integrates with the existing Knowledge Base to deliver instant, accurate, and sourced answers to CSRs.

Vector DB
Pinecone / FAISS

Embeddings
OpenAI / Sentence Transformers

LLM
GPT-4 / Claude

Pipeline
KB Content Ingestion

CSR Types Question

Semantic Search Engine

Knowledge Articles Retrieved

GenAI Summarizes & Generates

CSR Receives Answer + Source

GenAI in Action — Live Demo Flow

CSR QUESTION

"Can employees add dependents during Annual Enrollment?"

AI RESPONSE

Answer: Employees can add eligible dependents during the Annual Enrollment period. The change becomes effective on January 1 of the following year.

1. Log in to the Benefits Portal
2. Select Manage Dependents
3. Upload required documentation (if applicable)
4. Submit changes before the enrollment deadline

Source: Knowledge Article – "Dependent Eligibility and Enrollment Guidelines"

How GenAI Transforms the KB Experience

Intelligent Search

Traditional: Keyword matching only
With GenAI: Semantic understanding of intent

Example: 'change health plan' → modify benefits, update coverage, switch insurance plan

Content Summarization

Traditional: CSR reads 2,000-word SOPs
With GenAI: Instant 5-line actionable summary

Long articles become quick, structured answers for agents

Knowledge Gap Detection

AI logs all unanswered / low-confidence queries

KM team can identify: missing articles, outdated content, and frequently searched topics

Coverage Areas

5 mock knowledge base articles:
· Health Plan Enrollment
· Dependent Eligibility
· Coverage Changes
· Enrollment Deadlines
· Premium Deductions

The CSR Experience Transformation

✕ WITHOUT GENAI

- 📖 CSR reads 3+ articles manually
- ⏱ 2–3 minutes per search
- ⚠ High error rate & inconsistency
- 📉 Low confidence for new agents
- 🔍 Keyword-only search results
- 😩 Agents feel overwhelmed during peaks

✓ WITH GENAI

- ⚡ AI delivers answer instantly
- ⏱ Under 30 seconds per query
- ✓ Standardized, consistent answers
- 📈 New agents onboard faster
- 🧠 Semantic intent understanding
- ☺ Agents handle more calls with confidence

Measurable Value Delivered

Reduced Handle Time

CSRs spend less time searching and more time resolving customer issues.

↓ 40% AHT

Faster Onboarding

New agents ramp up quickly with AI-guided responses — no deep KB expertise needed.

Reduced ramp time

Consistent Answers

Every CSR delivers the same accurate information — reducing misinformation risk.

Standardized CX

KM Intelligence

Query logs surface gaps in the knowledge base for continuous improvement.

Data-driven KB

Thank You

Empowering CSRs with the right answers at the right time —
through Generative AI and intelligent knowledge retrieval.

Generative AI · RAG Architecture · Knowledge Management · Customer Service